

Problem 3

A complex number a is unimodular if $|a| = 1$. Show that a complex number a is unimodular if and only if it can be written as the quotient of two complex conjugate numbers.

Solution

We need to prove both directions of the equivalence.

Definition

A complex number a is **unimodular** if $|a| = 1$, meaning it lies on the unit circle.

Theorem Statement

To Prove: $|a| = 1$ if and only if there exist complex numbers z and $\bar{z}$ such that:

$$a = \frac{z}{\bar{z}}$$

where $\bar{z}$ is the complex conjugate of z (and $z \neq 0$).

Part 1: Forward Direction ($\Rightarrow$)

Given: $|a| = 1$

To Show: a can be written as $\frac{z}{\bar{z}}$ for some $z \neq 0$.

Proof

If $|a| = 1$, we can write a in polar form:

$$a = e^{i\theta} = \cos \theta + i \sin \theta$$

for some real θ .

Now, let's choose:

$$z = e^{i\theta/2}$$

Then:

$$\bar{z} = e^{-i\theta/2}$$

Computing the quotient:

$$\frac{z}{\bar{z}} = \frac{e^{i\theta/2}}{e^{-i\theta/2}} = e^{i\theta/2 - (-i\theta/2)} = e^{i\theta} = a$$

Therefore, $a = \frac{z}{\bar{z}}$ with $z = e^{i\theta/2}$.

Alternative Construction

Alternatively, for any a with $|a| = 1$, we can choose any non-zero complex number w and set:

$$z = a^{1/2}w$$

Then:

$$\bar{z} = \overline{a^{1/2}w} = \overline{a^{1/2}} \cdot \bar{w}$$

And:

$$\frac{z}{\bar{z}} = \frac{a^{1/2}w}{\overline{a^{1/2}w}} = \frac{a^{1/2}}{\overline{a^{1/2}}} \cdot \frac{w}{\bar{w}}$$

Since $|a| = 1$, we have $|a^{1/2}| = 1$, so $a^{1/2}$ is also unimodular.
 If we choose w such that $\frac{w}{\bar{w}} = 1$ (e.g., w real and positive), then:

$$\frac{z}{\bar{z}} = \frac{a^{1/2}}{\overline{a^{1/2}}}$$

Actually, the simplest approach is the first one shown.

Part 2: Reverse Direction ($\Leftarrow$)

Given: $a = \frac{z}{\bar{z}}$ for some $z \neq 0$

To Show: $|a| = 1$

Proof

Using the property $|w_1/w_2| = |w_1|/|w_2|$:

$$|a| = \left| \frac{z}{\bar{z}} \right| = \frac{|z|}{|\bar{z}|}$$

Since $|\bar{z}| = |z|$ (the modulus of a complex number equals the modulus of its conjugate):

$$|a| = \frac{|z|}{|z|} = 1$$

Therefore, a is unimodular.

Verification with Examples

Example 1

Let $a = i$ (which is unimodular since $|i| = 1$).

Choose $z = e^{i\pi/4} = \frac{1}{\sqrt{2}} + \frac{i}{\sqrt{2}}$.

Then:

$$\bar{z} = e^{-i\pi/4} = \frac{1}{\sqrt{2}} - \frac{i}{\sqrt{2}}$$

$$\frac{z}{\bar{z}} = \frac{e^{i\pi/4}}{e^{-i\pi/4}} = e^{i\pi/2} = i = a$$

Example 2

Let $a = -1$ (unimodular since $|-1| = 1$).

Choose $z = e^{i\pi/2} = i$.

Then:

$$\bar{z} = e^{-i\pi/2} = -i$$

$$\frac{z}{\bar{z}} = \frac{i}{-i} = -1 = a$$

Example 3 (Reverse Direction)

Let $z = 3 + 4i$. Then $\bar{z} = 3 - 4i$.

$$a = \frac{z}{\bar{z}} = \frac{3 + 4i}{3 - 4i}$$

Multiply numerator and denominator by the conjugate of the denominator:

$$= \frac{(3 + 4i)(3 + 4i)}{(3 - 4i)(3 + 4i)} = \frac{9 + 24i + 16i^2}{9 - 16i^2} = \frac{9 + 24i - 16}{9 + 16}$$

$$= \frac{-7 + 24i}{25}$$

Check modulus:

$$|a| = \frac{\sqrt{(-7)^2 + 24^2}}{25} = \frac{\sqrt{49 + 576}}{25} = \frac{\sqrt{625}}{25} = \frac{25}{25} = 1$$

Geometric Interpretation

The quotient $\frac{z}{\bar{z}}$ represents:

- The modulus: $|\frac{z}{\bar{z}}| = \frac{|z|}{|z|} = 1$ (always on unit circle)
- The argument: $\arg\left(\frac{z}{\bar{z}}\right) = \arg(z) - \arg(\bar{z}) = \arg(z) - (-\arg(z)) = 2\arg(z)$

So $\frac{z}{\bar{z}}$ is the point on the unit circle at angle $2\arg(z)$.

Final Answer

$$|a| = 1 \iff a = \frac{z}{\bar{z}} \text{ for some } z \in C^*$$

Proof Summary:

($\Rightarrow$) If $|a| = 1$, write $a = e^{i\theta}$ and choose $z = e^{i\theta/2}$. Then:

$$\frac{z}{\bar{z}} = \frac{e^{i\theta/2}}{e^{-i\theta/2}} = e^{i\theta} = a$$

($\Leftarrow$) If $a = \frac{z}{\bar{z}}$, then:

$$|a| = \left| \frac{z}{\bar{z}} \right| = \frac{|z|}{|\bar{z}|} = \frac{|z|}{|z|} = 1$$

Key Insight: The quotient of a complex number by its conjugate always has modulus 1, and conversely, every unimodular number can be represented this way.